**MES ABASAHEB GARWARE COLLEGE (AUTONOMOUS), PUNE**

**DEPARTMENT OF COMPUTER SCIENCE**

# PROJECT REPORT

## ON

# WEATHORA

## WEATHER DASHBOARD WITH API

**Submitted By**

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**Internship Organization**

Naviotech Solutions

**Project Allocation:** Web Development Project – 2026

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Finally, I am thankful to all those who helped me in completing the Weathora project successfully.

**With regards,**

**Sanika Sanjay Dhavale**

# INTRODUCTION

**Weathora – Weather Dashboard with API** is a web-based weather application developed to provide users with current weather information of different cities. Users can enter a city name and view its current weather details through a simple and user-friendly interface.

The application uses the **OpenWeather API** to fetch real-time weather information. It is developed using **Python, Flask, HTML, CSS and JavaScript**. **MongoDB** is used to store users' weather search history.

## I. EXISTING SYSTEM

In the traditional method, people often depend on television, newspapers or other weather applications to obtain weather information. Sometimes, finding specific city weather details quickly can be difficult.

Users may also need to visit different websites or applications to check weather conditions for different locations.

## II. NEED OF NEW SYSTEM

There is a need for a simple and convenient system that provides current weather information quickly. Users should be able to search for any city and view its weather details in one place.

Weathora provides a user-friendly platform where users can search for a city, view its current weather information and access their previous search history. It helps users obtain weather information easily through a web-based application.

# PROBLEM DEFINITION

Weather conditions can change frequently, so people need updated weather information to plan their daily activities. However, finding current weather details for a particular city may sometimes require visiting different websites or applications.

Users need a simple platform where they can search for a city and quickly view its current weather information. There is also a need to maintain a record of previously searched cities for easy reference.

Therefore, the problem is to develop a simple and user-friendly web-based Weather Dashboard that allows users to search for different cities, view current weather information using an API and access their previous search history.

# PROPOSED SYSTEM

Weathora is a web-based Weather Dashboard developed to provide current weather information in a simple and convenient way. The system allows users to enter a city name and retrieve its current weather details using the OpenWeather API.

The application is developed using Python and Flask for backend functionality, along with HTML, CSS and JavaScript for the user interface. MongoDB is used to store weather search history.

## I. EXPLANATION

The user enters a city name in the search box. The application sends a request to the OpenWeather API and retrieves the current weather information for that city. The fetched information is then displayed on the dashboard.

The system also stores search history in MongoDB, allowing users to view their previous weather searches. The application provides Home, Weather, History and About pages for easy navigation.

## II. METHODOLOGY USED

The project follows a simple development approach:

1. Designing the user interface using HTML and CSS.
2. Developing backend functionality using Python Flask.
3. Integrating the OpenWeather API to fetch current weather information.
4. Connecting MongoDB to store weather search history.
5. Testing the application to check its functionality.

# SCOPE OF THE SYSTEM

The scope of the Weathora project is to provide a simple and user-friendly web-based platform for checking current weather information of different cities.

The system provides the following facilities:

1. **City Search:** Users can enter the name of a city to search for its weather information.
2. **Current Weather Information:** The application displays current weather details retrieved from the OpenWeather API.
3. **Weather Dashboard:** Users can view weather information through a simple and attractive interface.
4. **Search History:** The system stores previous weather searches in MongoDB.
5. **History Page:** Users can view their previously searched weather information.
6. **About Page:** Provides information about the Weathora application.
7. **User-Friendly Interface:** The application provides easy navigation through its different pages.

The project focuses on displaying current weather information and maintaining search history. It can be further improved by adding additional weather-related features in the future.

# HARDWARE AND SOFTWARE REQUIREMENTS

**Hardware Requirements:**

• Processor : Intel(R) Core(TM) i7-10850H CPU @ 2.70GHz (2.71 GHz)

• RAM : 8.00 GB (7.61 GB usable)

• Storage : 96 GB of 477 GB used

• Operating System: Windows 11

**Software Requirements:**

• Frontend: HTML, CSS, JavaScript

• Backend: Python and Flask

• Database: MongoDB

• Weather API: OpenWeather API

• Development environment: Python/Flask development setup

• Browser: Chrome, Edge or another modern web browser

# FACT FINDING TECHNIQUES

The following fact-finding techniques were considered to understand the requirements of the Weathora system:

• **Observation:** Observed how users search for weather information through different websites and applications.

• **Document Study:** Referred to weather-related information and OpenWeather API documentation to understand the required weather data.

• **Requirement Analysis:** Identified the main functions required for city search, displaying current weather information and maintaining search history.

• **System Study:** Studied the working of weather applications to identify the basic features required for the Weathora project.

# FEASIBILITY STUDY

The feasibility study helps to determine whether the proposed Weathora system is practical and suitable for development. It considers operational, technical and economical aspects of the system.

## I. Operational Feasibility

Weathora provides a simple and user-friendly interface for users to search for a city and view its current weather information. Users can also access their previous weather searches through the History page. The system makes it convenient to obtain weather information through a web application.

## II. Technical Feasibility

The system is technically feasible because it uses commonly available technologies such as Python Flask, MongoDB, HTML, CSS and JavaScript. The OpenWeather API is used to retrieve current weather information. The application can be developed and operated on a standard computer with an internet connection.

## III. Economic Feasibility

Weathora is economical to develop because it uses freely available and open-source technologies such as Python, Flask and MongoDB. It does not require expensive hardware or software, which makes the overall development cost low.

# DATABASE DESIGNING

Weathora uses MongoDB as its database to store weather search history. MongoDB is a NoSQL database that stores data in the form of documents.

## I. DATABASE DETAILS

- Database Name: weathora
- Collection Name: search_history
- Database Type: NoSQL (MongoDB)

## II. COLLECTION STRUCTURE

The `search_history` collection stores the following fields:

1. **_id:** Unique identification number automatically generated for each record by MongoDB.
2. **guest_id:** Unique ID assigned to each browser session to maintain separate search history.
3. **city:** Name of the searched city.
4. **country:** Country code of the searched city.
5. **temperature:** Current temperature of the searched city.
6. **description:** Weather condition of the searched city.
7. **searched_at:** Date and time when the city was searched.

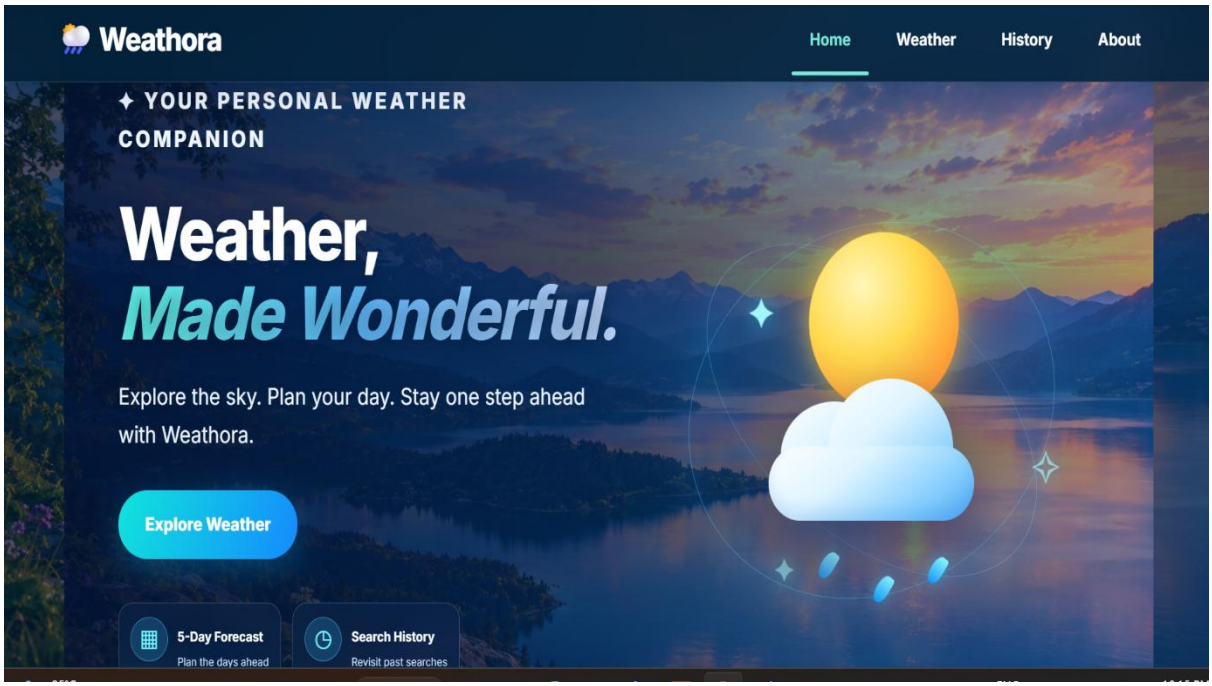
## III. DATABASE OPERATIONS

1. **Insert:** Stores weather search details in MongoDB.
2. **Find:** Retrieves the previous search records of the current browser session.
3. **Sort:** Displays the latest search records first.
4. **Delete:** Allows users to delete an individual search record.
5. **Clear History:** Allows users to delete all search records from their browser session.

The system uses MongoDB operations to maintain and retrieve weather search history efficiently.

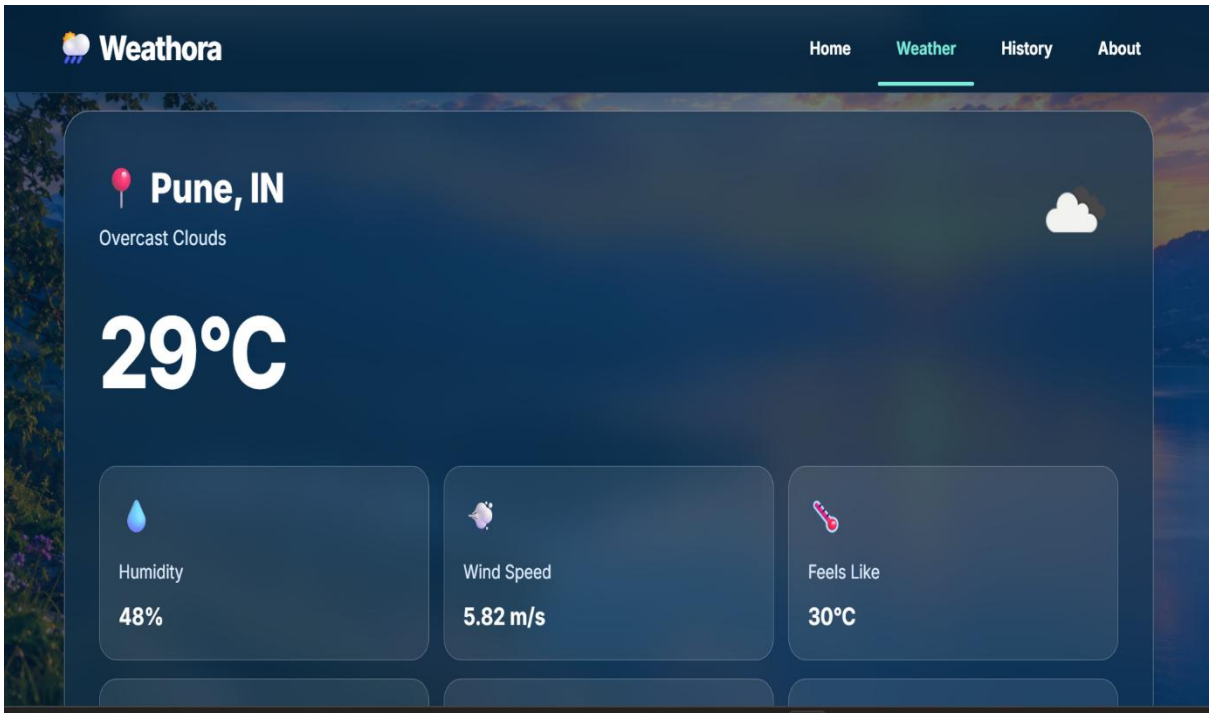
# SCREEN DESIGNING

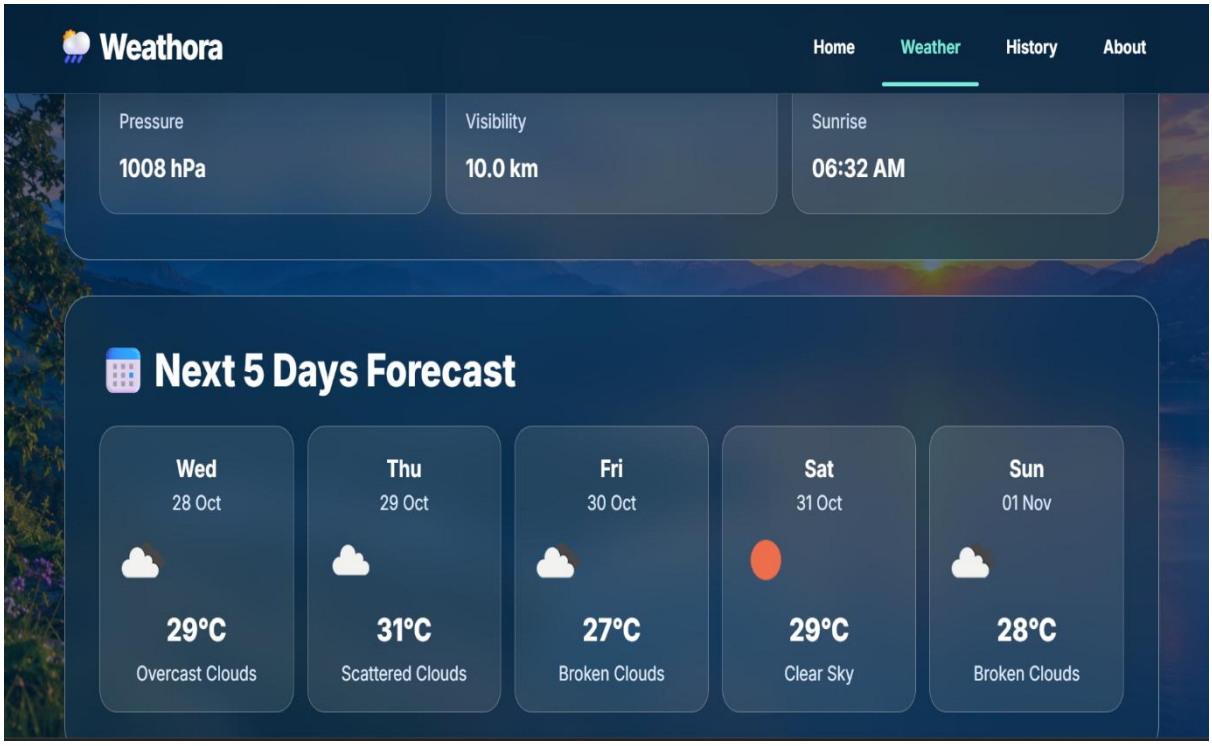
## Home Page:

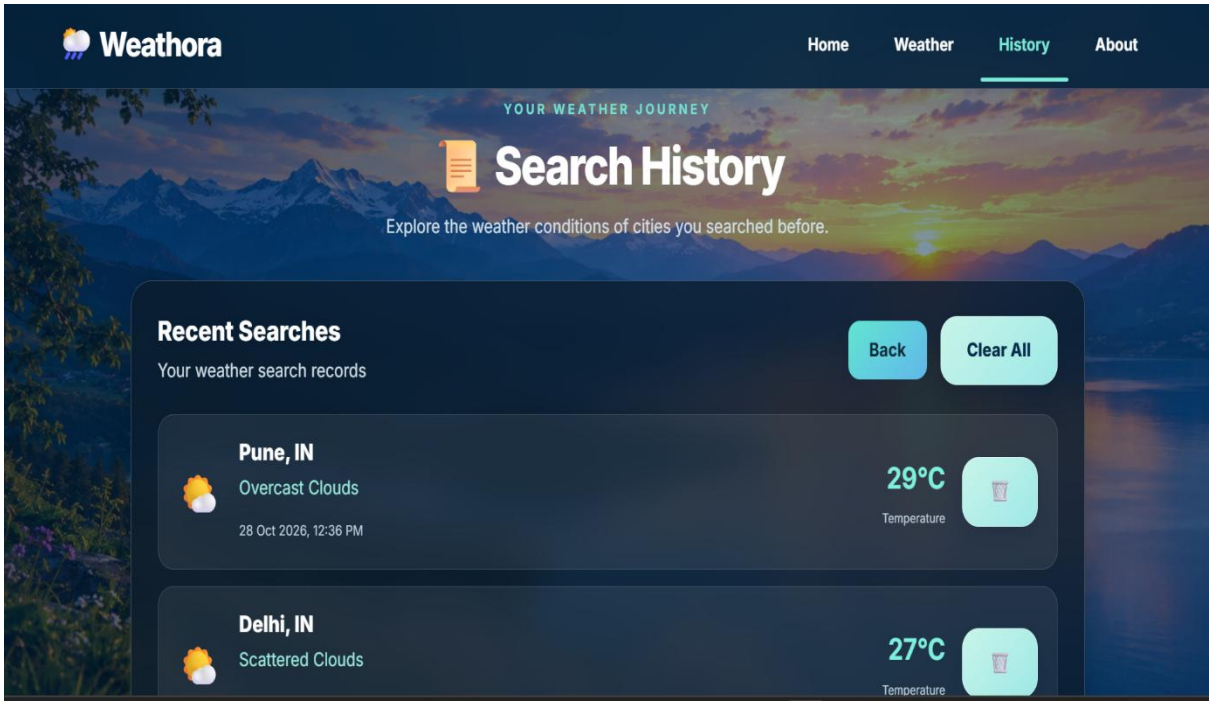



## Weather Forecast Page:



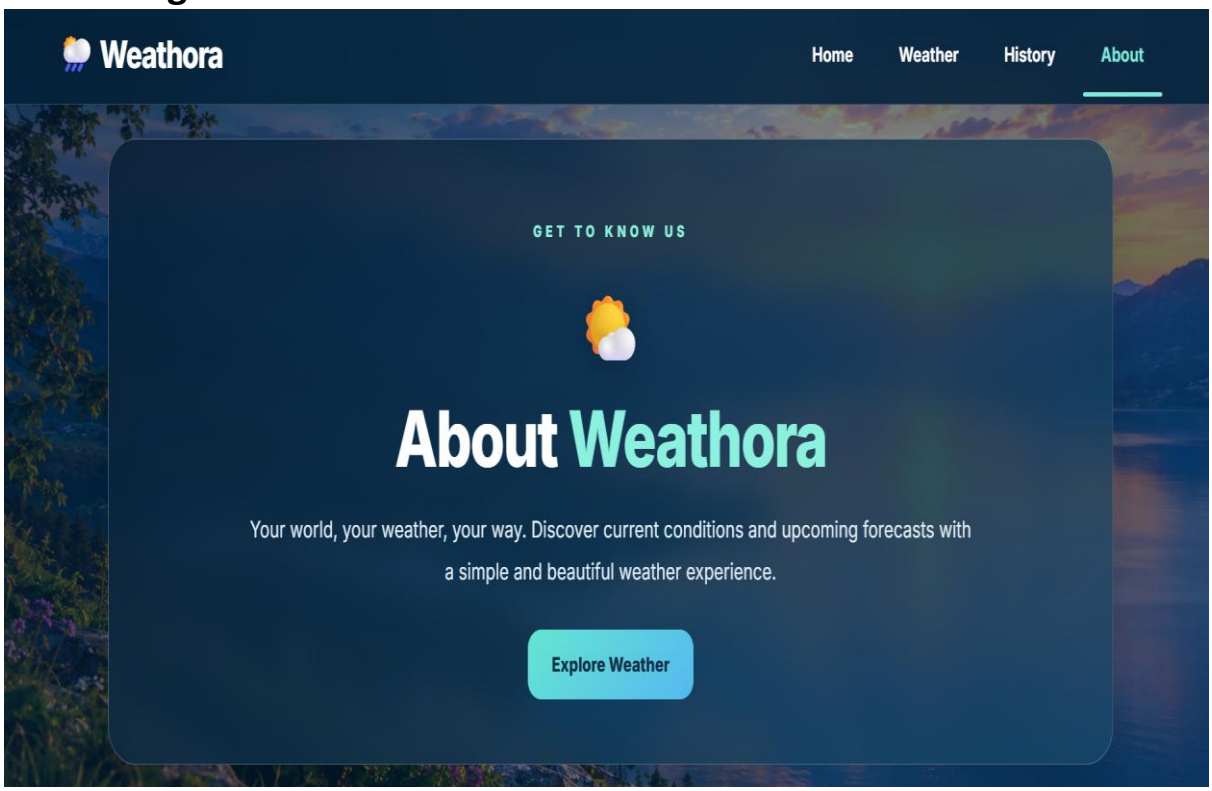

Weathora
Home
Weather
History
About
Pune, IN
Overcast Clouds
29°C
Humidity
48%
Wind Speed
5.82 m/s
Feels Like
30°C


Weathora
Home
Weather
History
About
Pressure
1008 hPa
Visibility
10.0 km
Sunrise
06:32 AM
Next 5 Days Forecast
Wed
28 Oct
29°C
Overcast Clouds
Thu
29 Oct
31°C
Scattered Clouds
Fri
30 Oct
27°C
Broken Clouds
Sat
31 Oct
29°C
Clear Sky
Sun
01 Nov
28°C
Broken Clouds




**Search History Page:**

**About Page:**



# TEST CASES DESIGN

Testing is an important part of the Weathora project. It helps to check whether the application functions correctly and provides the expected results.

The following test cases are considered for testing the Weathora application:

| Sr. No. | Test Case | Expected Result |
|---|---|---|
| 1 | Open the Home page | Home page should open successfully. |
| 2 | Search with a valid city name | Current weather information should be displayed. |
| 3 | Search with an invalid city name | City not found message should be displayed. |
| 4 | Check weather details | Temperature, humidity, wind speed and other details should be displayed. |
| 5 | Open the History page | Previously saved search records should be displayed. |
| 6 | Delete a search record | Selected search record should be deleted. |
| 7 | Clear search history | All search records of the current session should be deleted. |
| 8 | Open the About page | About page should open successfully. |

## II. TESTING CONCLUSION

The test cases help verify the main functions of the Weathora application, including city search, weather information display, search history management and page navigation.

# LIMITATIONS

Although Weathora provides a simple and convenient platform for checking weather information, it has certain limitations.

1. **Internet Connection:** An active internet connection is required to fetch weather information from the OpenWeather API.
2. **API Dependency:** The application depends on the OpenWeather API to retrieve weather data. If the API service is unavailable, weather information may not be displayed.
3. **API Key Requirement:** A valid OpenWeather API key is required for fetching weather information.
4. **City Search:** Users need to enter a valid city name to get the correct weather information.
5. **Search History:** Search history depends on MongoDB connectivity. If the database is unavailable, search records cannot be saved or retrieved.
6. **Limited Functionality:** The application mainly focuses on weather information and search history. It does not provide advanced weather analysis features.

These limitations can be reduced by adding more features and improving the application in future versions.

# CONCLUSION AND FUTURE ENHANCEMENTS

## I. CONCLUSION

Weathora – Weather Dashboard with API is a web-based application developed to provide current weather information of different cities. The application allows users to search for a city and view its weather details through a simple and user-friendly interface.

The project uses Python and Flask for backend development, HTML, CSS and JavaScript for frontend development, MongoDB for storing search history and OpenWeather API for retrieving weather information.

Through this project, practical knowledge of web development, API integration, database connectivity and application testing was gained. Overall, Weathora provides a convenient way to access weather information through a web application.

## II. FUTURE ENHANCEMENTS

The following features can be added to Weathora in future:

1. **Weather Alerts:** Notifications can be added for severe weather conditions.
2. **Weather Maps:** Interactive maps can be included to display weather conditions of different locations.
3. **Improved Forecast:** More detailed weather forecast information can be provided.
4. **Location Detection:** Automatic location detection can be added to show local weather information.
5. **Mobile Compatibility:** The application can be further improved for a better mobile user experience.

These enhancements can make Weathora more useful and provide additional weather-related features in future versions.

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